Team Number :	apmcm24200658
Problem Chosen :	C

2024 APMCM summary sheet

This study offers a comprehensive examination of future trends in the pet industry both in China and globally, utilizing various mathematical modeling techniques to forecast market developments. The research employs tools such as multiple linear regression, time series analysis, and game theory to investigate changes in pet populations, pet food consumption dynamics, and the effects of policy shifts on the market.

In response to question 1, we analyzed data from China's pet industry over the last five years, focusing on cat and dog population changes. Initially, time series analysis was employed to identify overall pet population trends, followed by multiple linear regression to evaluate how macroeconomic factors like GDP per capita, consumer spending on pets, and urbanization influence these populations. Our predictions indicate continued growth in cat numbers, while dog population growth may decelerate, signaling a shift in market demand.

For question 2, we analyzed the US and European markets, utilizing panel data models to assess the relationship between pet ownership and pet food market size across countries. By creating a vector autoregressive (VAR) model, we examined the dynamic interplay between market size and pet numbers, predicting a significant expansion of the global pet food market due to rising pet ownership.

In addressing question 3, we analyzed production and export data from China's pet food industry. Using the Holt-Winters time series model, we forecasted the industry's gross production value and applied multiple regression analysis to understand how various economic factors influence export trends. Our model also assesses the effects of global economic changes on China's pet food exports and anticipates future recovery and growth in export markets.

Regarding question 4, we explored the implications of new external economic policies, particularly tariff adjustments, on China's pet food sector. We developed static and dynamic game-theoretic models to simulate strategic interactions with major trading partners under the new policy environment. The analysis reveals that policies from key European and American countries significantly affect China's pet food exports, leading us to propose optimal strategies for enhancing the international competitiveness of China's pet food industry.

Finally, we evaluated the strengths and limitations of our model, providing scientific insights for the sustainable development of the pet industry, particularly in light of uncertainties in international trade policy, and offering guidance for strategic adjustments in China's pet food sector.

Keywords: time series, static and dynamic game theory, forecasting, Nash equilibrium analysis

Content

I、 Introduction

1. 1 Background

Since the early 21st century, China's pet industry has undergone significant growth driven by the gradual liberalization of pet-related policies. Pets have evolved from being simple household aids to vital emotional companions. Urbanization and demographic changes, such as an aging population and declining birth rates, have further propelled the industry's expansion, resulting in flourishing ancillary sectors. Increased consumer expenditure on pets has transformed purchasing habits, escalating the demand for pet food, supplies, veterinary care, and services.

Emerging in the 1990s, China's pet industry coincided with rising disposable incomes following economic reforms. The relaxation of governmental restrictions facilitated the entry of international brands, such as Royal Bear and Little Star, marking the sector’s initial growth phase. Although the rate of dog ownership has plateaued, other pet categories remain robust, and independent brands are capturing market share. The emergence of innovative retail models that blend online and offline sales is influencing the market, alongside advancements in scientific pet care and intelligent management solutions.

Globally, the pet industries in Europe and the United States continue to expand rapidly, holding a substantial portion of the global market. Nonetheless, China's pet sector encounters challenges from evolving domestic and international policies, including post-Trump tariff reforms, as well as fluctuating market demands and economic conditions.

This study aims to assess the development of China's pet industry over the past five years, taking into account global market dynamics and potential tariff implications. It seeks to establish a scientific framework for forecasting future trends and to provide strategic recommendations for sustainable growth in both China's and the global pet industry.

1. 2 Question Restatement

Leveraging the provided datasets and supplementary information gathered by our team, we will perform a comprehensive analysis of the pet industry’s development trends and market demands. In conjunction with the prevailing economic context, we will propose strategic recommendations for the advancement of the pet industry in China.

1.2.1Question 1:

This paper analyzes the development of China's pet industry in the past five years, especially the changes in the number of pet cats and dogs, and discusses the factors affecting the development of China's pet industry. Build a mathematical model to predict the development of China's pet industry in the next three years.

1.2.2 Question 2:

In recent years, the pet industry in Europe and the United States has witnessed substantial growth. Utilizing the data presented in Annex 2 and supplementary information collected by the team, an analysis of global pet industry trends by species is warranted. Furthermore, this study will examine the primary factors influencing the global pet industry's development and establish an appropriate mathematical model to forecast the demand for pet food over the next three years.

1.2.3Question 3:

Based on the data regarding China's pet food production and export value presented in Annex 3, along with the trends in global demand for pet food and China's developmental trajectory, please conduct an analysis of the current state of China's pet food industry and project the production and export outlook for the next three years, assuming the continuity of existing economic policies.

1.2.4Question 4:

The pet food sector in China will undoubtedly be influenced by the new economic policies implemented by European nations and the United States, including tariff regulations. Develop a suitable mathematical model, utilizing the provided data, supplementary information gathered, and the outcomes of the preceding analyses. Based on these computations, formulate a viable strategy for the sustainable advancement of China's pet food industry.

II、Problem analysis

2. 1 Analysis of Question 1

The aim of the analysis for question 1 is to evaluate the evolution of China's pet industry over the last five years and to forecast trends for the upcoming three years. This involves a preliminary examination of historical pet population data through descriptive statistics and visualization techniques to illustrate annual changes in the number of pet cats and dogs. Subsequently, multiple linear regression will be employed to identify significant economic and social determinants influencing pet ownership, including per capita GDP and pet-related expenditures. To enhance forecasting precision, a hybrid approach combining time series analysis (like the ARIMA model) with principal component regression (PCR) will be utilized, leveraging both the temporal trends and the variations explained by key influencing factors for predictions regarding the pet industry's growth over the next three years.

2. 2 Analysis of Question 2

Question 2 seeks to analyze the global development trends of the pet industry and forecast future demand for pet food. This will involve a thorough assessment utilizing pet population and market data from various countries over recent years. We will employ descriptive statistical analysis and visualization methods, including trend line charts, to illustrate the trends in pet numbers and the pet food market. Furthermore, a panel data regression model will be developed to examine the relationship between economic indicators such as per capita GDP and urbanization rates and the pet food market size. To enhance the accuracy of our predictions regarding market demand and the dynamic interactions among variables, we will also implement vector autoregressive (VAR) models for a deeper analysis.

2. 3 Analysis of Question 3

In addressing Question 3, we will perform a comprehensive analysis and projection of the production and export metrics within China's pet food sector. Initially, a time series analysis will be conducted on pertinent data, incorporating seasonal decomposition and trend examination to elucidate the primary patterns in production and export figures. Following this, we will integrate external economic variables (such as shifts in international trade policies and the global economic landscape) alongside internal industry metrics (including domestic production capacity and market demand). By employing cointegration and error correction models (ECM), we will simultaneously assess the long-term equilibrium and short-term dynamics, facilitating precise forecasts for future production and export activities.

2. 4 Analysis of Question 4

Question 4 will examine international trade policies, focusing on the effects of tariff modifications on China's pet food sector. We will employ a game theory framework to analyze and forecast effective strategies for the industry. Specifically, a bilateral game model will simulate the strategic interactions between China and key trading partners, such as the United States, to evaluate optimal export pricing decisions across various tariff scenarios. By determining the Nash equilibrium, we will identify the most advantageous strategy for China's pet food industry within a specified tariff regime. Furthermore, dynamic game analysis will be utilized to offer model-driven strategic recommendations for industry leaders, taking into account temporal strategy adaptations.

III、 Model assumptions

(1) Assuming that the pet market is in a state of supply and demand balance, the market price can naturally adjust the supply and demand relationship of pets, and there is no significant market imbalance;

(2) It is assumed that the macroeconomic indicators (such as GDP growth rate and household consumption level) will change steadily during the analysis period, and there will be no sudden economic crisis or extreme economic fluctuations;

(3) It is assumed that pet food production technology and efficiency remain constant and production costs and conditions are stable during the forecast period.

(4) It is assumed that all the data obtained and used are accurate and can truly reflect the actual situation of the market and industry, and there is no measurement error or missing data.

(5) Assuming that in the short term, domestic and foreign tax policies, trade policies and relevant laws have not changed, the short-term impact of external policy adjustments on the industry should be appropriately considered.

(6) It is assumed that the exchange rate involved in international trade remains stable in the short term and does not fluctuate significantly.

IV、 Description of the symbol

symbol	illustrate
β_x	Model regression coefficient
ò	Error term
M	Target variable
T_t	Time series prediction trend value
R_t	Multiple regression model predicted value
d	The highest order of polynomial
GDP	Gross Domestic Product

V、 Problem 1: Establishment and solution of the model

5. 1 Data analysis

First of all, a detailed analysis of the provided data on the number of pet cats and dogs is carried out, analyzing characteristics such as trends, growth rates, etc.

Table 1 Changes in the number of cats and dogs in China in different years

Year	The number of pet cats（ten thousand）	The number of pet dogs（ten thousand）
2019	4412	5503
2020	4862	5222
2021	5806	5429
2022	6536	5119
2023	6980	5175

In order to provide a clearer picture of the evolution of pet populations over time, our team has completed a histogram of the number of pet cats and dogs by year, as shown in Figure 2. Next, we will analyze the changes in the number of cats and dogs in detail.

As illustrated in Figure 2, the population of pet cats has experienced a consistent upward trajectory, increasing from 44.12 million in 2019 to 69.8 million in 2023, representing an approximate growth of 58%. In contrast, the number of pet dogs exhibited fluctuations during the same period, peaking at approximately 55.03 million in 2019 before declining between 2020 and 2022. Although the pet dog population rebounded to around 51.75 million in 2023, it remains below the 2019 levels.

The number of pet cats is showing a clear upward trend, while the number of pet dogs is showing a slight downward trend. By 2023, the number of pet cats has surpassed that of pet dogs, a phenomenon that reflects the growing number of families who prefer cats as pets.

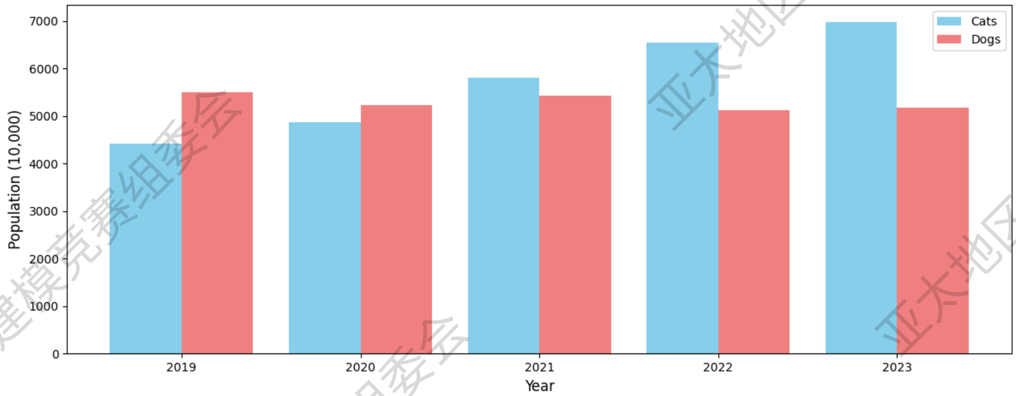

Fig.2 The number of pet cats and dogs changes over time

In order to further analyze the quantitative changes of the two, the growth rate of the two is calculated as follows:

$$Growth\ rate = \frac{Current\ year\ quantity - The\ previous\ year\ quantity}{The\ previous\ year\ quantity} \times 100\% \quad (1)$$

The growth trend of pet cats and pet dogs over time is shown in Figure 3. The growth of cats and dogs will be analyzed separately below:

For pet cats:

• In 2020, the growth rate was about 10%, marking the beginning of an accelerated increase in the number of pet cats.

• In 2021, the growth rate rose significantly to 19%, reaching its highest point, reflecting a sharp increase in demand for cats during the year.

• In 2022, growth slowed to 12.5%, but remained at a double-digit level.

• In 2023, the growth rate slowed further to 6.8%, and although the pace of growth continues to slow, market demand remains positive.

Overall, the number of pet cats is growing rapidly, but the rate of growth is gradually slowing.

For pet dogs:

• In 2020, the growth rate was about -5%, indicating a decrease in the number of pet dogs, possibly due to declining market demand or being limited by feeding policies.

• In 2021, the growth rate returned to 4%, indicating a recovery in demand for pet dogs.

• In 2022, the growth rate fell again to -6%, which may be related to external factors such as policy and the economy.

• In 2023, the growth rate rebounded to 1%, indicating that the demand for pet dogs is starting to stabilize.

Overall, the growth rate of pet dogs fluctuates greatly, showing a downward trend overall, but there are signs of stabilization in recent years.

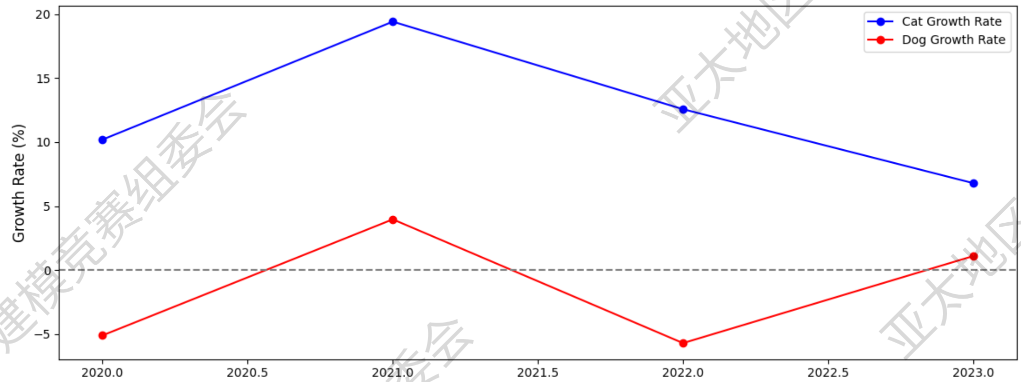

Fig. 3 Growth rate of pet cats and dogs over time

5. 2 Analysis of the development factors of China's pet industry

5.2.1 Data Collection

Through literature research, it was determined that the following factors have an impact on the development of the pet industry, as shown in Figure 4.

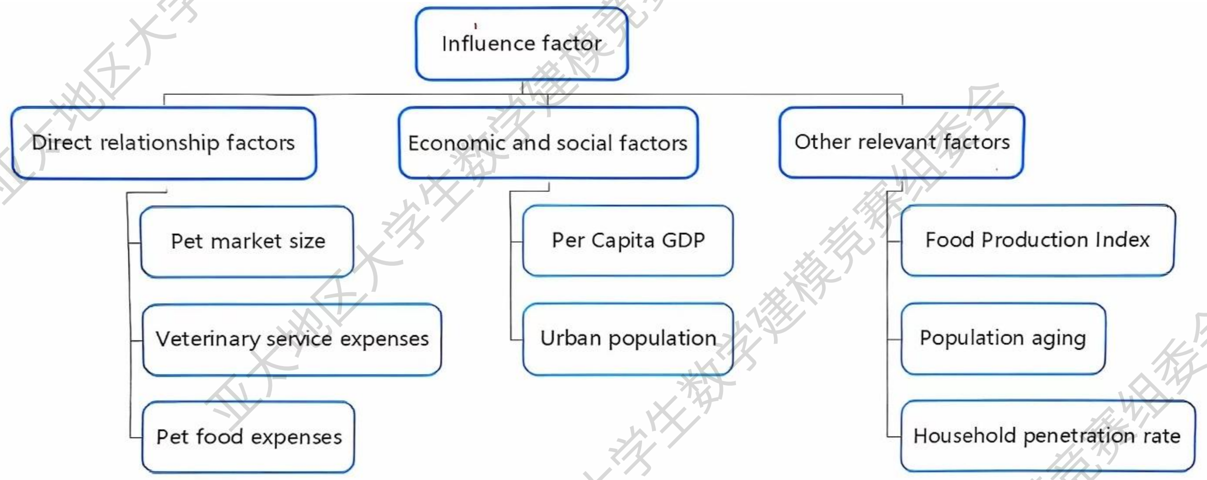

Figure 4 Indicator chart

5.2.2 Analysis of controlling factors

First, we will evaluate the strength of the linear relationship between the two independent variables, the number of pet cats and pet dogs, and the target variable (i.e., the number of cats or dogs) through correlation analysis. Next, we will use a ridge regression model to quantify the specific effect of each independent variable on the target variable, and in the process, solve possible multicollinearity problems. Finally, we will use LASSO regression to identify and screen out the independent variables that have the most significant impact on the target variables. Through this series of analyses, we can more accurately understand the relationship between pet numbers and identify the key factors that influence the target variables.

(1)Correlation analysis:

We will perform a quantitative analysis of the correlation between variables to evaluate the degree of linear correlation between them. Since the attachment data provided are numerical data, we will use the Pearson correlation coefficient method to measure the correlation between the variables. This approach gives us an accurate measure of the linear relationship between variables.

$$\rho_{X,Y}=\frac{\operatorname{cov}(X,Y)}{\sigma_X\sigma_Y}=\frac{E\left[\left(X-\mu_X\right)\left(Y-\mu_Y\right)\right]}{\sigma_X\sigma_Y} \tag{2}$$

The above equation is about the definition of the overall correlation coefficient, and the Greek symbol is often used ρ as the representative symbol. Then, the covariance and standard deviation of the sample are estimated to obtain the Pearson correlation coefficient, which is often represented by the English letter r

$$r=\frac{\sum_{i=1}^{n}\left(X_i-\bar{X}\right)\left(Y_i-\bar{Y}\right)}{\sqrt{\sum_{i=1}^{n}\left(X_i-\bar{X}\right)^2}\sqrt{\sum_{i=1}^{n}\left(Y_i-\bar{Y}\right)^2}} \tag{3}$$

r can also be estimated from the X_i standard fractional mean of the (,) sample points to obtain an equivalent expression to the above equation: Y_i

$$r=\frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma_X}\right)\left(\frac{Y_i-\bar{Y}}{\sigma_Y}\right) \tag{4}$$

$$\rho=1-\frac{6\sum d_i^2}{n\left(n^2-1\right)} \tag{5}$$

(2)Ridge return

Ridge regression is an improved technique of traditional linear regression, especially suitable for processing datasets with multicollinearity. In this method, the L2 norm is introduced into the loss function as a penalty term, and the regression coefficient is constrained to reduce the complexity of the model. This not only helps to alleviate the instability of parameter estimation caused by the high correlation between dependent variables, but also enhances the prediction performance of the model on new data.

$$\min_\beta\left\{\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}\beta_j^2\right\} \tag{6}$$

(3) LASSO regression

LASSO (Least Absolute Shrinkage and Selection Operator) regression is a regularization technique of linear regression, which constrains the model parameters by introducing the L1 norm as a penalty term in the loss function. Unlike ridge regression, LASSO is able to generate sparse solutions, which means that some of the coefficients are reduced to zero, thus achieving feature selection. This approach not only helps to improve the simplicity of the model, but also enhances its generalization ability.

The function takes the form of:

$$\min_{\beta}\left\{\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}|\beta_j|\right\} \tag{7}$$

(4) Analyze the results

Pet cat population analysis

(1) The pet market size and pet food expenditure had the highest positive correlation with the number of cats, indicating that these factors were the main driving factors for the change of cat population. There was a weak negative correlation between total fertility and the number of cats.

(2) The regression coefficient of pet food expenditure on the number of cats was higher, indicating that it had the greatest impact on the number of cats. Spending on veterinary services also has a significant impact on cat populations.

(3) The important factors screened out include pet food expenses and veterinary service expenses.

Pet dog population analysis

(1) The size of the pet market has a significant positive correlation with the number of dogs. Pet food spending is also strongly correlated with the number of dogs.

(2) Pet food expenditure and pet market size are the main contributors to the number of dogs. Other variables, such as China's pet household penetration rate, have a weaker impact on dog populations.

(3) The size of the pet market is the most important variable to be screened out.

Note: The horizontal coordinates of the following four bar charts are The size of pet market、Expenditure for pet food、Expenditure on Veterinary Services、Chinese pet household penetration rate、total fertility rate、Index number of food production、per capital GDP、urban population, from left to right

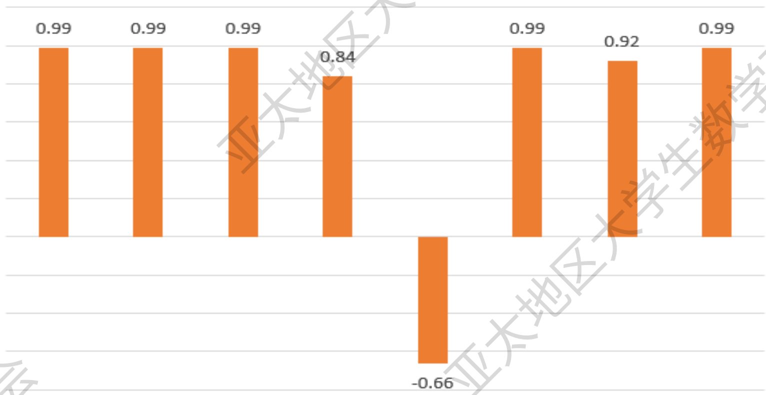

Fig.5 Correlation between each trait and the number of cats

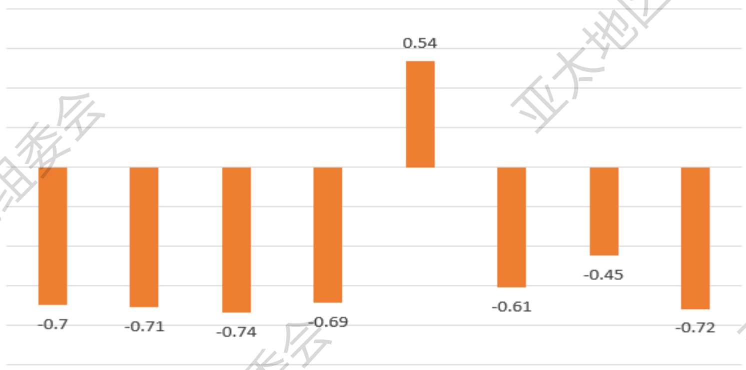

Fig.6 Correlation between each feature and the number of dogs

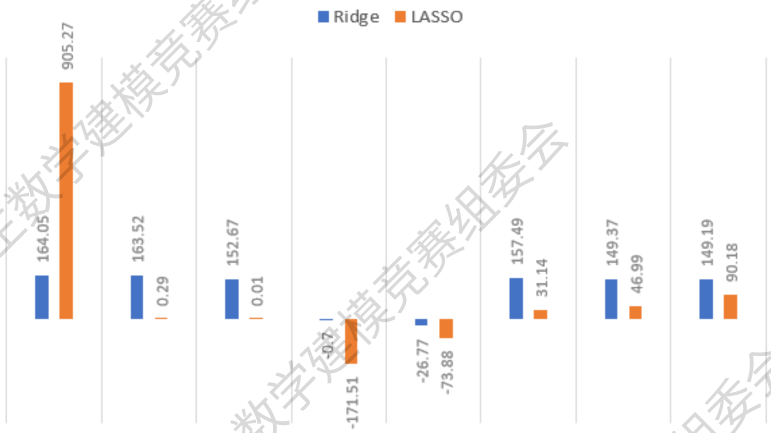

Fig.7 Regression coefficients of each trait and the number of cats

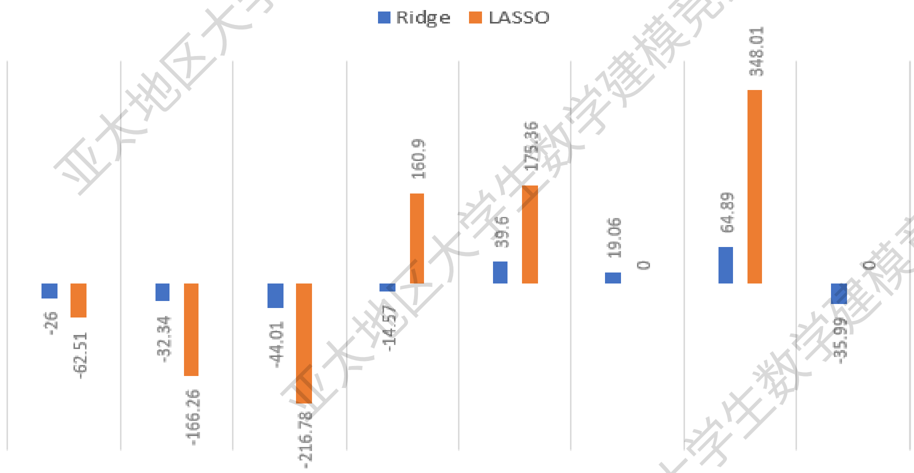

Fig.8 Regression coefficients of each trait and the number of dogs

5. 3 Forecast for the development of China's pet industry

5.3.1 Time series forecasting

Polynomial regression is an extended form of linear regression that describes the nonlinear relationship between dependent and independent variables by adding a higher-order term. Polynomial transformations of years can more effectively fit nonlinear trends that may exist in a time series.

Let the year be x and the target value (e.g. number of cats or dogs) be y. The model of polynomial regression is:

$$Correlation\ index = \frac{Covariance\ (X,Y)}{\sigma x \cdot \sigma y} \tag{8}$$

β —The regression coefficient of the model

x^d —The d-power of the year, which represents the higher order term in the polynomial

ò—Error term

The predicted value can be expressed as:

$$\hat{y} = \sum_{i=0}^{d} \beta_i x^i \tag{9}$$

where x: year and d: the highest order of the polynomial.

5.3.2 Random forest regression prediction

Random forest is a popular ensemble learning algorithm that builds multiple decision tree models based on the Bagging (Bootstrap Aggregating) method. In a random forest, each decision tree is a weak learner that is independently trained to make predictions about the sample. When it is necessary to classify or regress a new sample, the random forest collects the predictions of all decision trees and decides the final output by majority voting. This approach improves the accuracy of predictions because it reduces the risk of overfitting and takes advantage of the diversity of multiple models.

Information entropy is a measure of uncertainty in the amount of information, which reflects the diversity of different outcomes that an event may have. The lower the value of information entropy, the lower the uncertainty of the information. The calculation equation is as follows:

$$H(X) = -\sum_{i=1}^{n} P(X=i)\log_2 P(X=i) \tag{10}$$

Where: the probability of $P(X=i)$ a random variable $X=i$.

Conditional information entropy refers to the uncertainty of the random variable X given the random variable Y. The conditional entropy and information gain equations are as follows:

Conditional entropy: refers to the uncertainty exhibited by the random variable X when the random variable Y is known.

$$H(X|Y=v) = -\sum_{i=1}^{n} P(X=i|Y=v)\log_2 P(X=i|Y=v) \tag{11}$$

Information gain: refers to the degree to which the uncertainty of information is reduced under certain conditions.

$$I(X,Y) = H(X) - H(X|Y) \tag{12}$$

The Gini index reflects the probability that a randomly selected sample will be misclassified in a sample set. A smaller Gini index indicates a lower probability of misclassification, i.e., a higher purity of the sample (0 when all samples in the set belong to the same category). The calculation equation is as follows:

$$Gnini(p) = \sum_{k=1}^{K} p_k(1-p_k) = 1 - \sum_{k=1}^{K} {p_k}^2 \tag{13}$$

Random forest (RF) is a model that integrates multiple decision trees and can be used for both classification tasks and regression analysis. A schematic diagram of a decision tree and a random forest is shown in Figure 9.

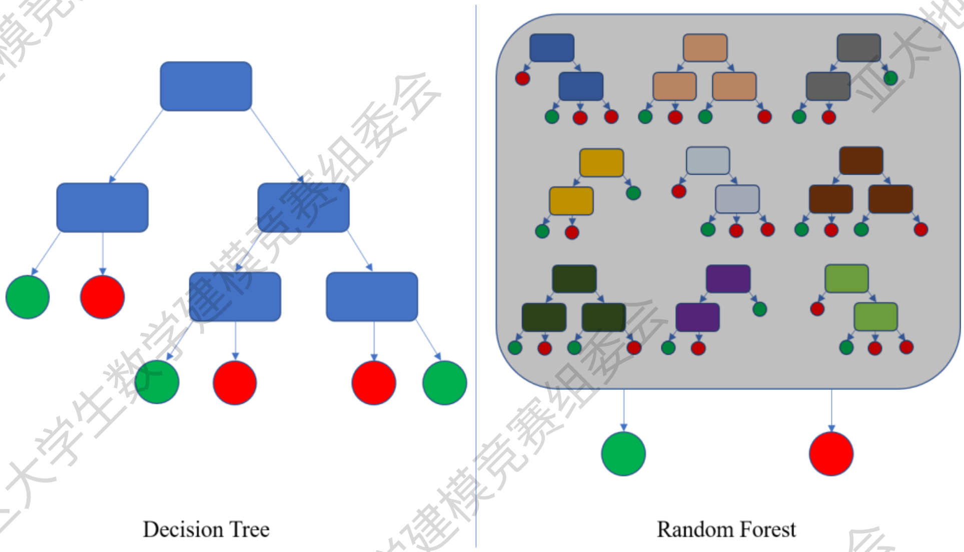

Fig.9 Schematic diagram of decision tree and random forest

For M trees, the prediction result for each tree is , then the final prediction value is:

$$\hat{y} = \frac{1}{M}\sum_{m=1}^{M} f_m(X) \tag{14}$$

5.3.3 Hybrid predictive models

The hybrid forecasting method combines time series forecasting and regression forecasting of the main controlling factors, and obtains the final prediction results in the form of weighted averages. Adjusting the weights controls the relative importance of the time series forecast and the master factor forecast in the overall forecast. $\hat{y}_{\text{hybrid}}$ The equation for calculating the mixed prediction is:

$$\hat{y}_{\text{hybrid}} = w\hat{y}_{\text{mc}} + (1-w)\hat{y}_{\text{ts}} \tag{15}$$

The prediction results are shown in Figures 10 and 11.

(1) Prediction results of the number of cats

Actual Trends: From 2019 to 2023, the number of pet cats continues to rise, reflecting the continued growth of the pet cat market size and demand.

Forecast trends: From 2024 to 2026, the number of cats will continue to show an upward trend, and although it continues to grow, the pace may slow down.

This trend is consistent with the natural characteristics of cat population growth, and the prediction results of the model show an appropriate nonlinear growth pattern. The predicted number of cats is reasonable, indicating that the model successfully identifies and reflects the growth trends presented in the historical data. The number of cats is expected to maintain a steady growth trajectory over the next three years, suggesting that the pet cat market has broad growth potential.

(2) Prediction results of the number of dogs

Actual trends: From 2019 to 2023, the number of dogs showed significant volatility, with large increases and decreases during the period, reflecting a high level of instability in historical data. Since 2022, the number of dogs has gradually stabilized.

Predict trends

Projections show that the number of dogs is expected to remain relatively stable between 2024 and 2026, with a limited range of fluctuations. This trend may be due to the influence of volatility in historical data and the tendency of models to smooth out these fluctuations.

Time Series Trend Analysis: Overall, the model is relatively accurate in predicting cat populations, and its growth trends are consistent with historical data. Forecasts for dog populations, on the other hand, smooth out fluctuations and show a possible stable trend in the future. The hybrid model combines time series with major influencing factors, reasonably models the nonlinear trend in historical data, and comprehensively considers the impact of multiple main controlling factors (such as pet market size, food expenditure, etc.).

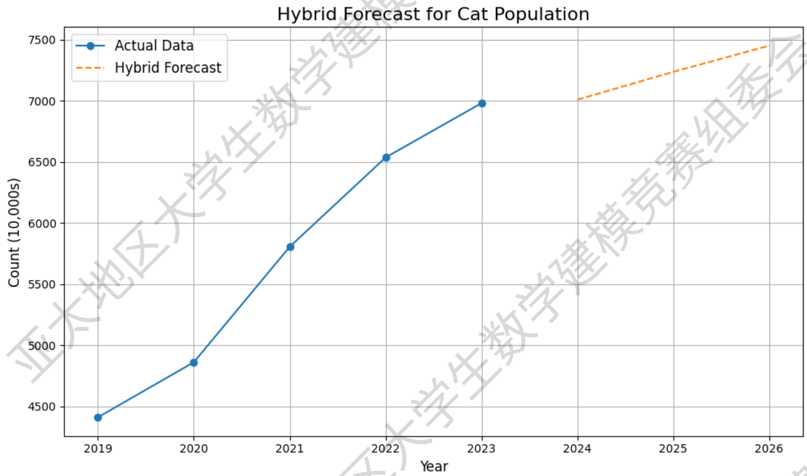

Fig.10 Prediction of cat population

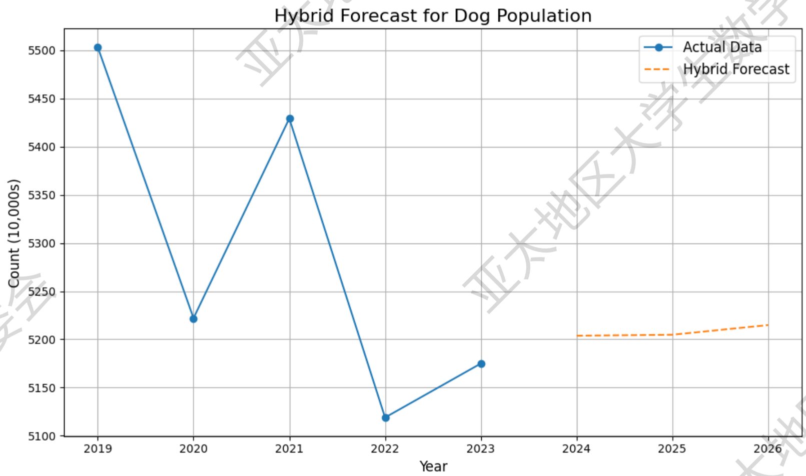

Fig.11 Prediction of dog population

VI、 Problem 2: Establishment and solution of the model

6. 1 Data analysis

Start with a trend analysis. The results are shown in Figure 12~Figure 15.

(1) Analysis of the number and growth trend of cats

In terms of quantitative trends:

The number of pet cats in the United States is visibly higher than in France and Germany.

From 2019 to 2023, there is an overall downward trend, indicating that the number of pet cats in the United States is gradually decreasing.

The number of pet cats in France and Germany is relatively close, with France having slightly more.

The data shows that the number of pet cats in France is on the rise, while the number of pet cats in Germany is more stable, but there is also a slight increase.

In terms of growth rate:

The growth rate of pet cats in the United States experienced significant fluctuations, with a sharp decline in 2020 and a slight recovery in 2021, but then declining again, culminating in a modest increase in 2023.

The instability of the growth rate also reflects the instability of the pet cat market in the United States.

The rate of growth of pet cats in France continues to rise steadily, indicating that the number of people owning cats is on the rise.

The growth rate of pet cats in Germany tends to be stable and does not fluctuate much, but the growth rate is slightly slower than that of pet cats in France.

(2) Analysis of dog numbers and growth trends

In terms of quantitative trends:

The number of pet dogs in the United States is also significantly higher than in France and Germany, and the number of pet dogs has been relatively stable between 2019 and 2023, with a slight decrease.

The number of pet dogs in France, although slightly lower than in Germany, is slowly increasing.

The number of pet dogs in Germany is close to the number of pet dogs in France, but slightly more. However, compared to France, the number of pet dogs in Germany has decreased slightly after 2022.

In terms of growth rate:

The growth rate of pet dogs in the United States fluctuates significantly, but the overall trend is declining, especially after 2022, and the negative growth is even more obvious.

After 2022, the number of pet dogs in France has increased significantly, reaching a high level in 2023, indicating that the dog breeding market in France has great potential.

The growth rate of the pet dog population in Germany has stabilized, and the overall level is low, indicating that the dog breeding market in Germany has reached a relatively saturated state.

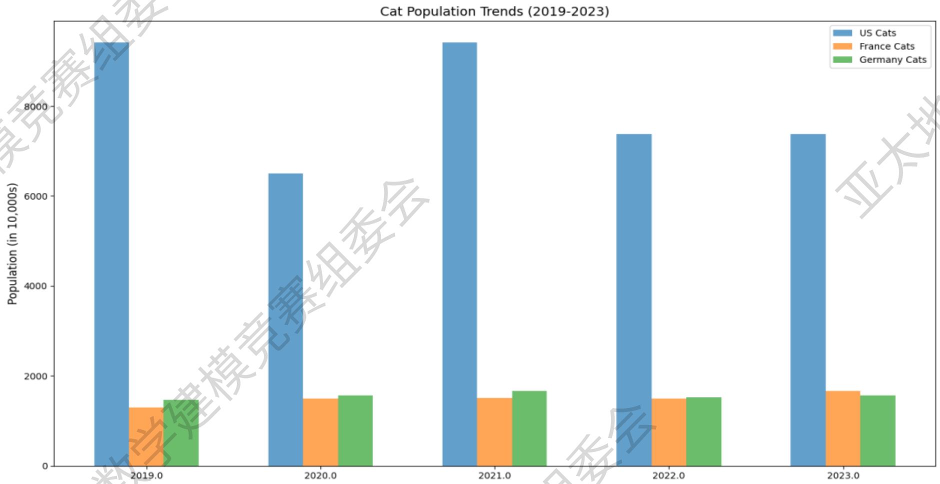

Fig. 12 Trends in cat populations by country

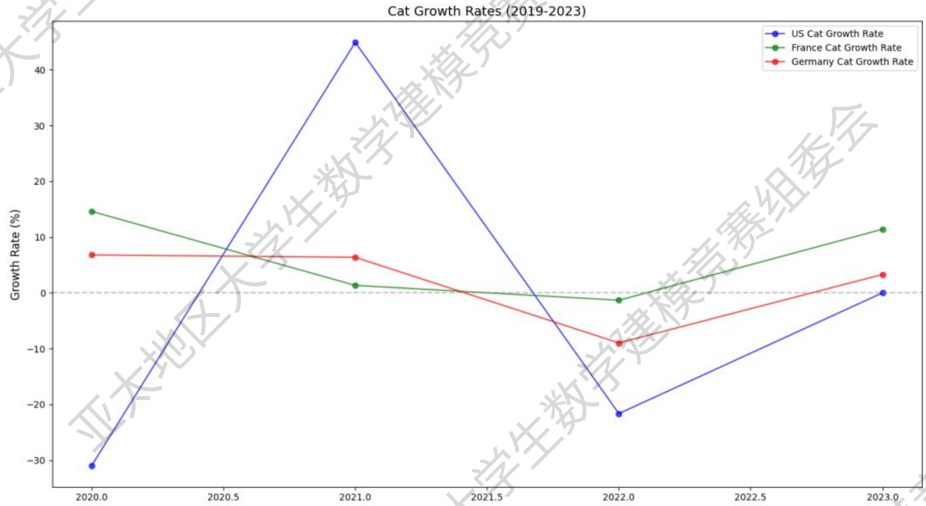

Fig. 13 Trends in the growth rate of cat populations by country

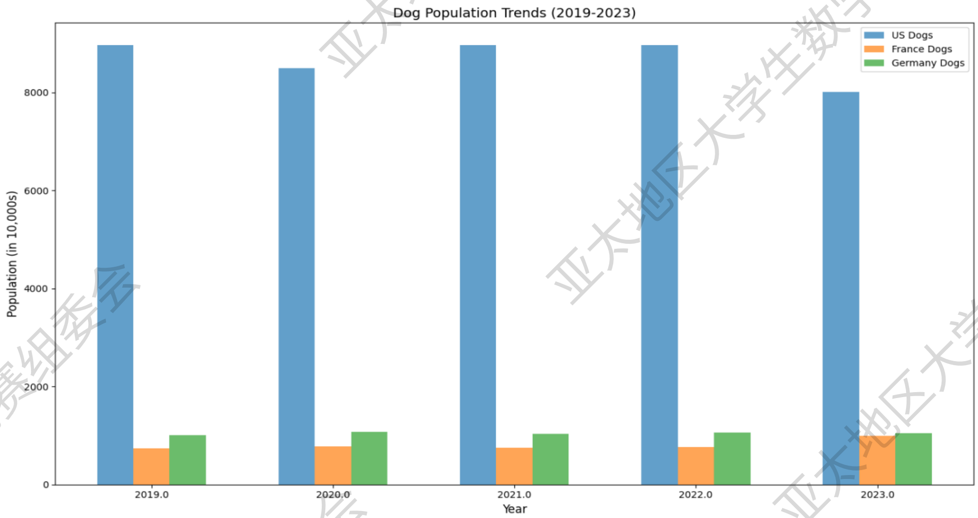

Fig. 14 Trends in dog populations by country

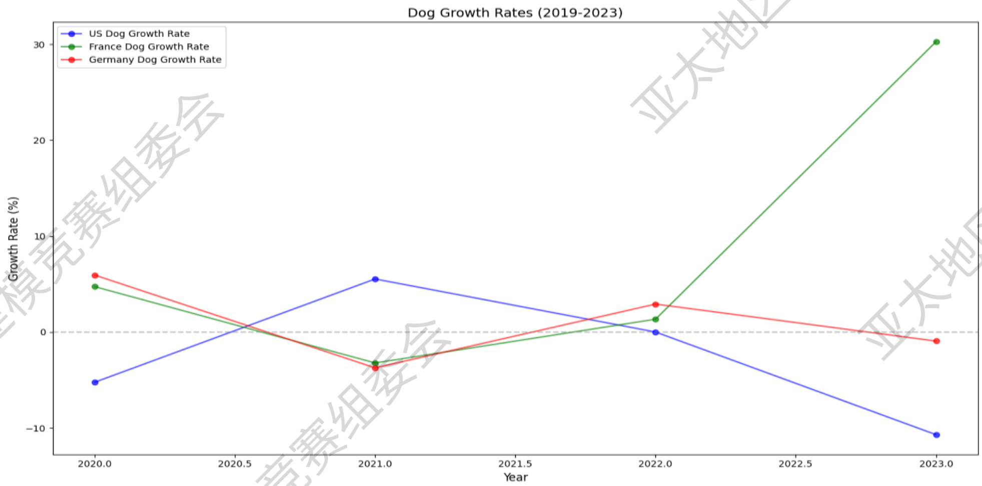

Figure 15 Trends in the growth rate of dog populations by country

6. 2 Forecast of pet food demand in other countries

6.2.1 Data Collection

(1) The number of pets

Annual statistics on pet cats and dogs by country provide insights into yearly trends in pet ownership, aiding in the prediction of overall pet food demand.

(2) Relevant data of the pet food market

Pet Food Industry Market Size (USD 100 million): This indicator reflects the overall consumption level of the pet food market and is a direct measure of market demand.

Annual Pet Food Expenditure ($100 million): This data provides the total amount of pet food spent each year and helps estimate the average food consumption per pet.

Total Pet Food Exports (USD 100 million): This indicator reveals the demand for pet food in the global market and is of great significance for global market analysis.

(3) Other relevant indicators of the pet industry

Pet Market Size (USD 100 million): This indicator maps the overall trend of pet-related consumption, which is indirectly related to the demand for pet food, reflecting the breadth and depth of economic activity in the pet industry.

Veterinary Services Expenditure (US$ 100 million): reflects pet owners' willingness to pay for pet consumption, which indirectly determines their spending on pet food.

(4) Socio-economic and demographic factors

Pet Household Penetration: This indicator reflects the popularity of pet ownership, which can indirectly infer the potential size of the pet market.

GDP per capita (USD): reflects the level of a country's national economy and is highly correlated with the ability to spend on pet food.

Urban population: The level of urbanization may promote pet keeping habits, which indirectly affects the demand for pet food.

6.2.2 Feature construction, selection, and correlation analysis

(1) Feature construction

For more efficient and accurate modeling and prediction, we construct the following new features:

Total number of pets

$$P_{\text{total}} = P_{\text{cat}} + P_{\text{dog}} \tag{16}$$

Per capita spending on pet food

$$E_{\text{avg}} = \frac{M}{P_{\text{total}}} \tag{17}$$

Quantify the effect of features on the target variable by calculating the correlation coefficient between the variables:

$$Correlation\ index = \frac{Covariance\ （X, Y）}{\sigma x \cdot \sigma y} \tag{18}$$

(2) Feature selection

A total of 10 feature variables were obtained. In order to reduce redundancy and improve the efficiency and interpretability of the model, the following methods will be used to filter the independent variables.

- Correlation analysis of features

The correlation between features and target variables (market size) is analyzed by calculating the Pearson correlation coefficient, and a heat map is generated using Python. The results are shown in Figure 16.

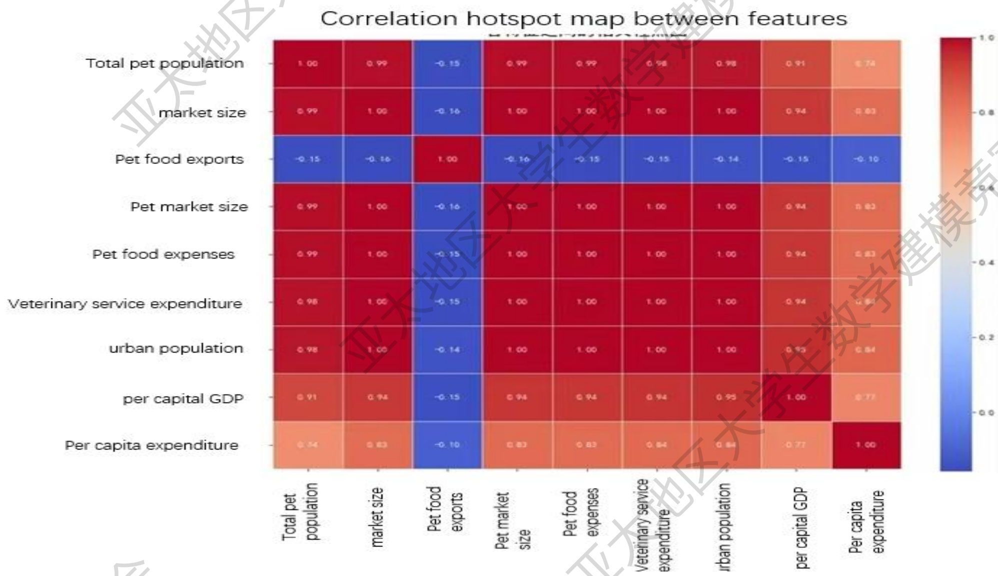

Fig.16 Correlation analysis results

The correlation coefficient between the total number of pets and the market size is extremely high at 0.96, indicating that the number of pets is a major factor driving the growth of the pet food market.

The correlation coefficient of per capita pet food expenditure is 0.78, which indicates that it has a certain importance in measuring the consumption power of the pet food market.

Although the correlation of urban population is not high, it still deserves in-depth analysis as a basic role of consumption.

- Eliminate redundant features

  Number of cats and dogs: Comprehensively reflected by the total number of pets, eliminated.

  Characteristics such as pet food export value, pet market size, and veterinary service expenditure: the direct correlation with market size is low; There may be collinearity between the data, culling.

- The final selection feature

  Based on our team's analysis of the data, we will select the following 4 characteristics as independent variables:

argument	Select a reason
Total number of pets	Indicates the pet base of the market and reflects the potential size of the market.
Urban population	Provide a reference to the number of consumers, the larger the population size, the higher the market demand.
GDP per capita	Reflecting the level of economic development of the country, the more developed the economy, the larger the pet food market.
Pet food per capita	The higher the pet owner's willingness to spend on pets, which plays a decisive role in increasing the market size.

6.2.3 Model construction and solving

(1) Multiple linear regression model

Multiple linear regression equations:

$$M = \beta_0 + \beta_1 P_{\text{total}} + \beta_2 P_{\text{urban}} + \beta_3 GDP_{\text{percapita}} + \beta_4 E_{\text{avg}} + \epsilon \tag{19}$$

The regression results are as follows:

United States

$$M = -120 + 0.021 P_{\text{total}} + 0.005 P_{\text{urban}} + 0.63 GDP_{\text{percapita}} + 140 E_{\text{avg}} \tag{20}$$

R2=0.985， MSE=1.3

France

$$M = -55 + 0.017 P_{\text{total}} + 0.002 P_{\text{urban}} + 0.51 GDP_{\text{percapita}} + 115 E_{\text{avg}} \tag{21}$$

R2=0.976， MSE=0.6

Germany

$$M = -62 + 0.019 P_{\text{total}} + 0.004 P_{\text{urban}} + 0.54 GDP_{\text{percapita}} + 130 E_{\text{avg}} \tag{22}$$

R2=0.981， MSE=0.8

(2) Time series model

Autoregressive (AR):

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \varepsilon_t \tag{23}$$

Differential (I): Eliminates non-stationarity of the data.

Moving Average (MA):

$$X_t = \mu + \theta_1 \varepsilon_{t-1} + \cdots \tag{24}$$

Figure 17 illustrates that the projected market size in the United States is anticipated to surpass that of France and Germany significantly, underscoring its greater market potential and consumption capacity. Furthermore, each country's market size is expected to experience modest growth by 2026 relative to 2024.

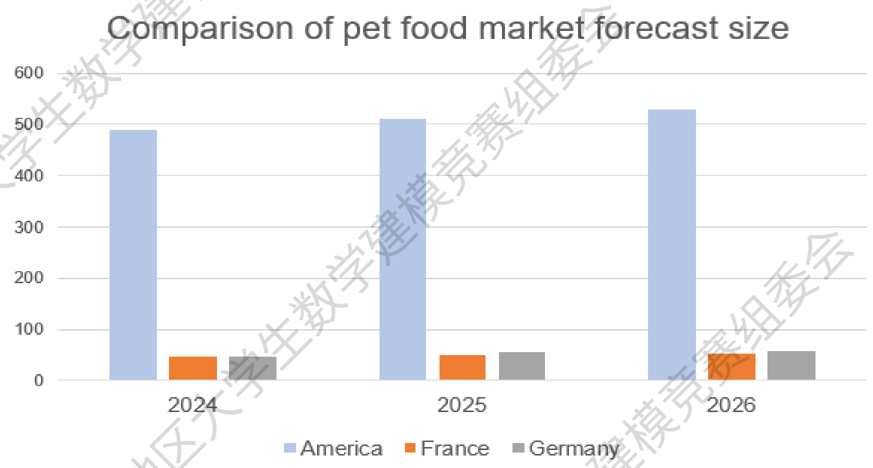

Figure 17 Market size forecast and comparison by country

VII、 Problem 3: Establishment and solution of the model

7. 1 Analysis of the development of China's pet food industry

Draw line charts and analyze trends.

(1) The trend of China's total output value of pet food

Analysis of the data of the total output value of pet food (RMB):

From 2019 to 2023, the total output value of the industry increased from 44.07 billion yuan to 279.3 billion yuan.

The growth rate reached 533.5%, with an average annual compound growth rate of about 56%.

This indicates that the industry has experienced rapid expansion over the past five years, especially in 2022 and 2023, with particularly significant increases, which may be affected by the expansion of the market size or other policy factors.

(2) The trend of China's total pet food exports

An observation of the total export value of pet food (USD) shows that from 2019 to 2023, the total export value decreased from US$15.41 billion to US$980 million, and then gradually recovered to US$3.96 billion.

There have been significant fluctuations along the way, especially in 2020, which experienced significant declines.

Possible reasons include the direct impact of changes in the external economic environment (e.g., the spread of the epidemic, adjustments in international trade policies) on exports.

(3) Changes in market demand in China's pet industry

Based on the total number of pets (millions) and pet food spending:

In 2019, the total number of pets was 99.8 million, and by 2023 it has increased to 132.02 million.

During this period, spending on pet food rose from US$15.1 billion to US$20.3 billion, with an average annual growth rate of about 7.5%.

This suggests that the growth in the number of pets is directly contributing to the demand in the domestic market, while consumer demand for the quality of pet food may also increase.

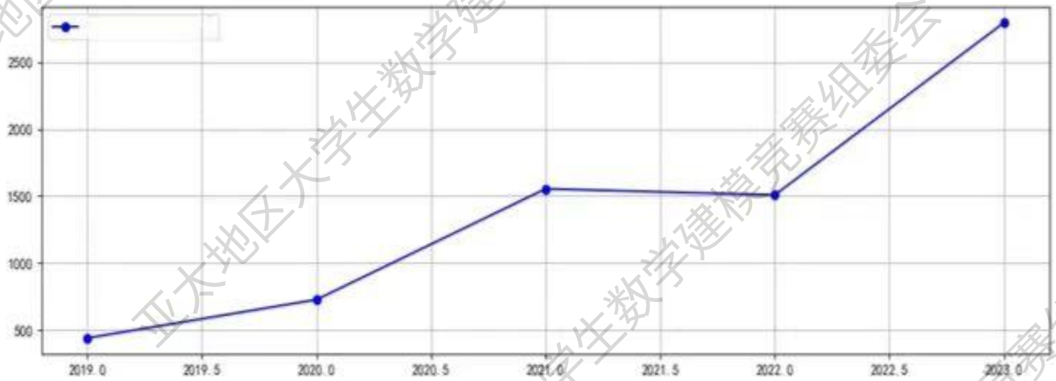

Figure 18 China's total output value of pet food (RMB, 100 million)

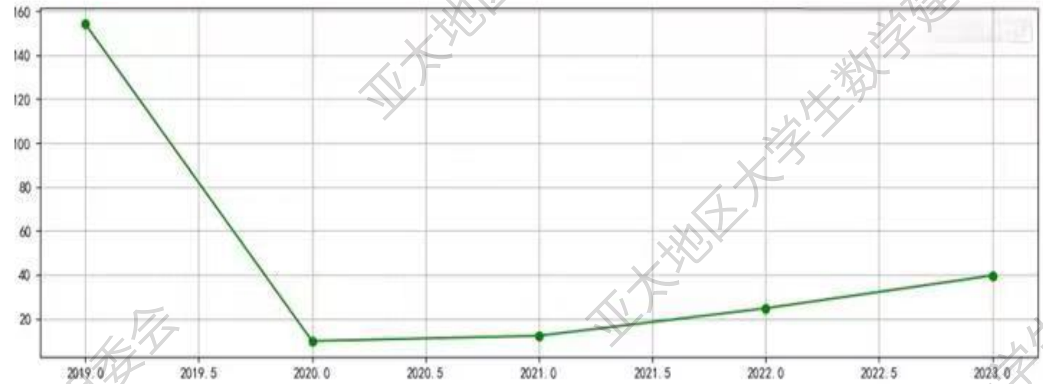

Figure 19 China's total pet food exports (US$, 100 million).

7. 2 Forecast for the future development of China's pet food industry

7.2.1 Data Collection

(1) Data related to the number of pets: the number of cats and dogs (unit: 10,000): the

basic market size used to analyze the demand for pet food.

Total number of pets (unit: millions): A direct reflection of the overall pet market size.

(2) Relevant data of pet food industry:

China's Pet Food Gross Output Value (RMB): Used to analyze the trend of the industry's production capacity.

China's total pet food exports (USD): directly reflects changes in the export market.

(3) Pet market related data:

Pet Market Size (in USD billion): Helps assess changes in the overall size of the industry.

Pet Food Expenditure (in US$100 million): Changes in market demand by segment.

(4) Economic and demographic factors (indirect effects):

Urban and rural population: reflects the geographical structure of the consumer market.

Food Production Index: Assesses the potential impact of overall domestic food production capacity on the pet food industry.

GDP per capita and final consumption expenditure of households: reflect the role of economic level and consumption power in driving market demand.

7.2.2 Data processing

(1) Description of variables

Target variable: China's total pet food output value (gross domestic product, unit: RMB).

China's total pet food exports in US dollars.

Influencing factor (independent variable): number of pets (millions).

Pet market size (USD billion).

Pet food spending (US$ billion).

Urban population (persons).

GDP per capita (USD).

Food Production Index (2014-2016 = 100).

(2) Data standardization

To prevent the influence of different units on the prediction results, we normalize the independent variables:

$$X' = \frac{X - \mu}{\sigma} \tag{25}$$

(3) Feature selection

Correlation Analysis:

The Pearson correlation coefficient between the target variable and the independent variable was calculated, and the characteristics with significant correlation between the two were screened out.

Multicollinearity detection:

The variance expansion factor (VIF) was used to evaluate multicollinearity between independent variables for the purpose of removing the characteristics of VIF>10.

Figure 20 shows the relevance of each feature to China's pet food industry. It is not difficult to see that:

(1) Pet number, pet market size, pet food expenditure: There is a strong positive correlation between these three indicators and GDP (0.95, 0.96, 0.95), which indicates that the rise of these variables usually leads to an increase in GDP. These factors may directly reflect the increase in domestic market demand.

(2) Total export value: There is a significant negative correlation between these factors and total export value (-0.59, -0.51, -0.52), which indicates that with the continuous expansion of the domestic market, the increase in domestic consumption may lead to a decrease in foreign exports.

(3) Urban population and per capita GDP: These two indicators show a significant positive correlation (0.94 and 0.82, respectively), indicating that the level of economic development and the process of urbanization may be the main factors driving production growth. However, the negative impact on the total value of exports is also significant.

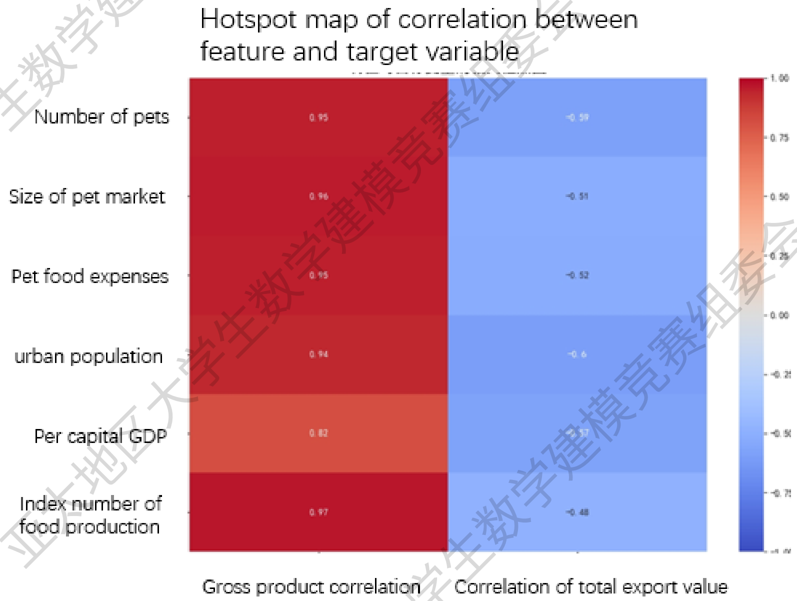

Fig.20 Heat map of the correlation between features and target variables

In the end, the characteristics chosen by our team are as follows:

Number of pets (millions). Pet market size (USD billion). Food Production Index (2014-2016 = 100).

7.2.3 Model building and solving

(1) Holt-Winters model

A Holt-Winters model in a time series model based on the target variable, and step-by-step break down the trend portions of the data:

Trending Items:

$$T_t = \alpha Y_t + (1-\alpha)(T_{t-1} + B_{t-1}) \tag{26}$$

Smoothing items

$$L_t = \beta(T_t - T_{t-1}) + (1-\beta)B_{t-1} \tag{27}$$

Predicted value

$$F\{t+m\} = Lt + mBt \tag{28}$$

(2) Multiple regression analysis

Use the regression model to fit the time series residual portions:

$$R_t = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon_t \tag{29}$$

(3) Hybrid model

The final forecast consists of two parts: the time series trend and the multiple regression residuals:

$$Yt = Tt + Rt \tag{30}$$

Figure 21 illustrates the time-series trend of China's real pet food production fitting to the Holt-Winters method. The solid line represents the actual annual data point, while the dotted line represents the predicted trend line. As can be seen from the figure, the model reflects the growth trend of GDP well, suggesting that it may continue to grow in the future.

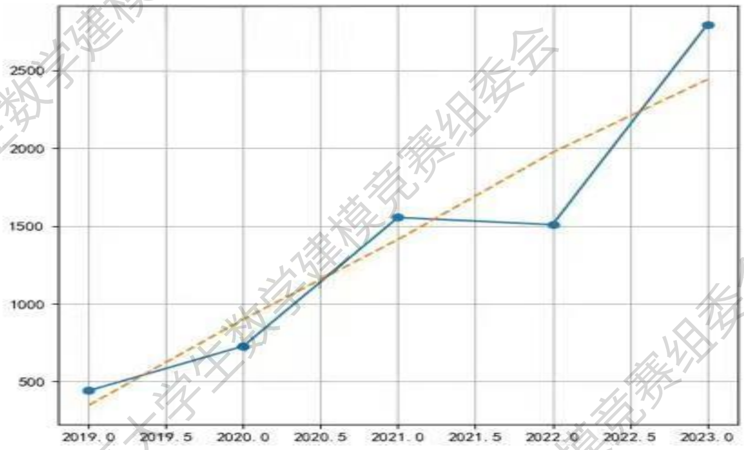

Fig.21 Time series trend of GDP (RMB, 100 million)

Blue: Real GDP ； Yellow: Holt-Winters trend

Similar to GDP, Figure 22 shows the actual data on the total value of exports and their trends. Although the total value of exports has declined since 2019, it has gradually recovered and shown growth since 2021, and the model predicts that this growth will continue in the future.

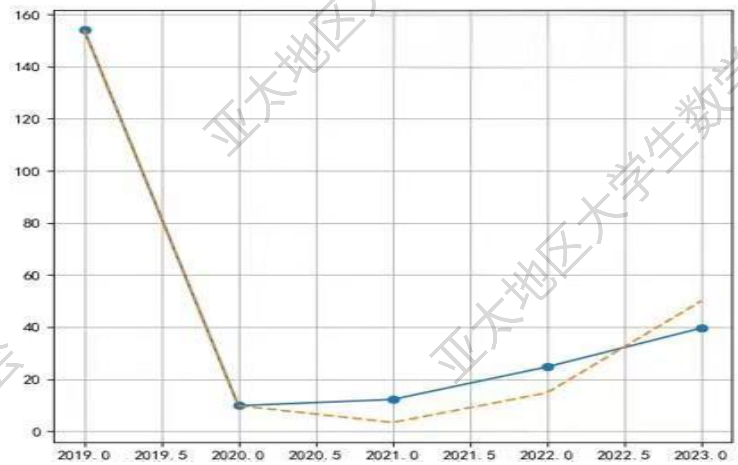

Fig. 22 Time series trend of total export value (US$, 100 million)

Blue: Real GDP ； Yellow: Optimized Holt-Winters trend

Figures 23 and 24 illustrate future trends in the pet food industry, with solid lines representing historical actual data and dotted lines indicating forecasted future values. Forecasts of GDP suggest that GDP will grow significantly over the next few years, especially after 2023, which may be closely related to a significant increase in market demand.

Forecast of total exports: Forecasts of total exports show signs of recovery and are expected to return to pre-2019 levels and continue to grow in the following years.

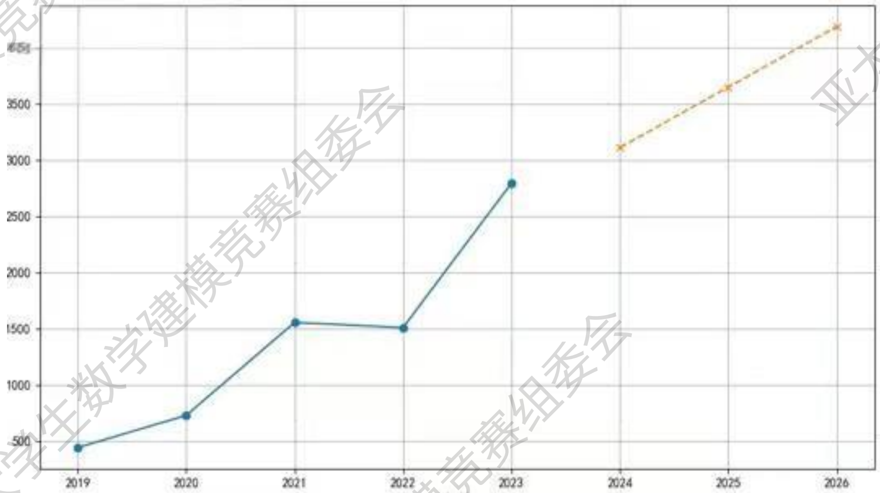

Fig. 23 Comparison of actual and forecasted GDP (RMB, 100 million yuan)

Blue (Real GDP); Yellow (Forecasted GDP)

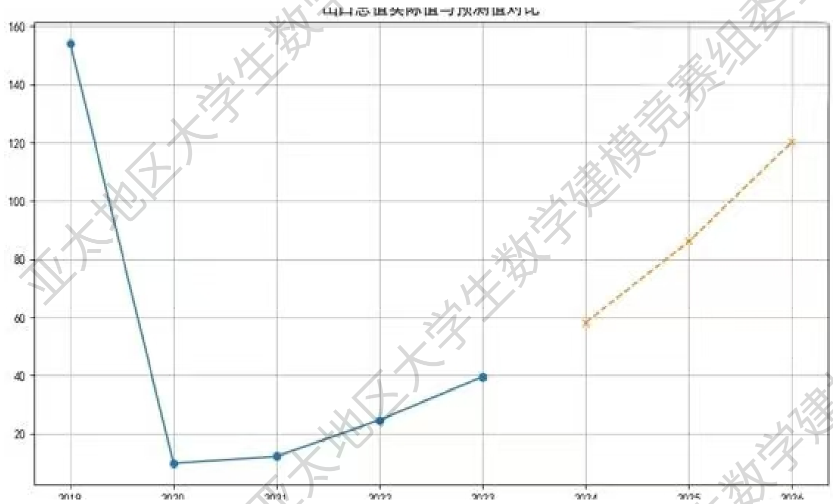

Figure 24 Comparison of actual and forecasted total exports (US$, 100 million)

Blue (Actual Total Exports) ; Yellow (Forecasted Total Exports)

VIII、 Problem 4: Establishment and solution of the model

Our team will conduct research using a static game method to simulate the strategic competition between China and the United States in the pet food market, construct a revenue matrix and solve the Nash equilibrium. At the same time, in order to analyze the dynamic adjustment of the strategy, the team members came up with the idea of introducing a dynamic game model.

8. 1 Assumptions and variable definitions

There are only two major players in the market for this model: China (Player 1) and the United States (Player 2).

The strategies held by both sides are to adjust export prices and maintain existing prices, respectively.

The revenue function will be affected by a variety of factors such as market share, tariff policy, and production costs.

Other countries and regions (e.g., Southeast Asia) have less impact on the pet food market and currently do not have a large share of the international market, so they are not included in the game analysis.

SC, SA: Market share in China and the United States.

PC, PA: Export Prices from China and the United States.

CC, CA: Production Costs in China and the United States.

T: U.S. tariff rate on China (25%Now).

πC, πA: the function of returns between China and the United States.

8. 2 Static game model

8.2.1 Gain function definition

The return function is defined as follows (after joining both China and the United States):

$$\pi_C = (P_C - C_C - T) \cdot S_C, \pi_A = (P_A - C_A) \cdot S_A \tag{31}$$

The relationship between market share and price can be approximated as:

$$S_C = \frac{1}{1+e^{\alpha(P_C - P_A)}}, S_A = 1 - S_C \tag{32}$$

Among them, α>0: market price sensitivity coefficient, which indicates the degree of impact of price differences on market share.

8.2.2 Policy space

China-Owned Strategy: Choosing Whether to Reduce Prices:

PC=P1 (maintain the current price).

PC=P2 (reduced price).

U.S.-Held Strategy: Choosing Whether to Raise Tariffs Further:

T = 25% (maintain existing tariffs).

T = 30% (tariff increase).

8.2.3 Earnings matrix

Combined with the above definitions, the benefit matrix is constructed:

	United States: Maintenance tariffs (T=25%)	United States: Tariff increase (T=30%)
China: Maintenance Price (PC=P1)	$\pi 11$, $\pi 12$	$\pi 13$, $\pi 14$
China: Maintenance Price	$\pi 21$, $\pi 22$	$\pi 23$, $\pi 24$

(PC=P2)		

8.2.4 Dynamic game model

Introduce a two-stage model of dynamic games:

Phase 1: The U.S. chooses whether or not to adjust the tariffs.

Phase 2: China chooses whether to lower prices based on the tariff adjustments given by the United States.

In this dynamic game, we will use the backward method to solve:

In Phase 2, China chooses the best price under the known tariff strategy:

$$\arg\max_{P_C} \pi_C \tag{33}$$

In Phase 1, the U.S. chooses a tariff strategy to maximize its own benefits:

$$\arg\max_{T} \pi_A \tag{34}$$

8. 3 Solving process and results

Using the above-mentioned return function and strategy space, the Nash equilibrium in the static game and the perfect equilibrium of the sub-game in the dynamic game are solved step by step.

The solution steps are as follows:

(1) Parameters

α=1.5: price sensitivity coefficient.

P C = [50,45], P A = 55 (USD).

C = 30, C A = 35 (USD).

T=25%,30%。

(2) Calculation of the Earnings Matrix:

The parameters are substituted into the return function to calculate the return matrix.

(3) Nash equilibrium solving:

Seek a combination of strategies that will make the best results for both strategies.

(4) Using dynamic game to solve:

Use the backward method to determine the optimal strategy for each stage.

The results of the static game, as shown in Figures 25 and 26.

From China's earnings matrix, it can be seen that:

When China's prices remain high (PC=50), China's earnings remain high regardless of the U.S. tariff policy. Specifically, if the tariff rate is 25% (T=0.25), China's gain is 7.50, while when the tariff rate rises to 30% (T=0.30), the gain decreases slightly to 5.00.

When China lowered the price to 45 (PC=45), there was a noticeable drop in earnings. At T=0.25, the return is 3.75, while at T=0.30, the return is further reduced to 1.50.

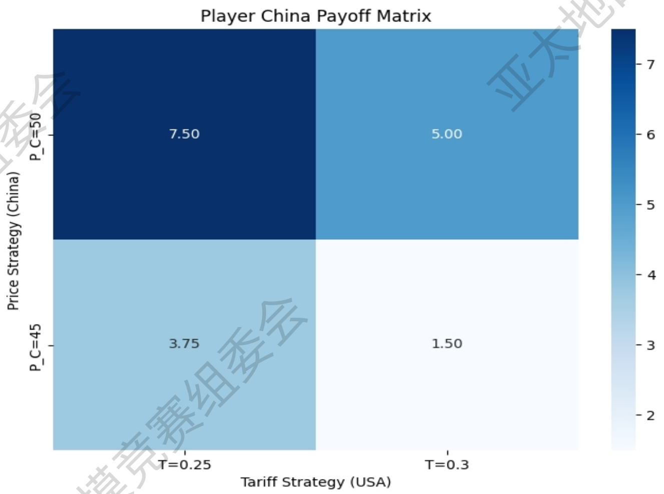

Figure 25 China's revenue matrix

The gains in the United States are relatively small and almost negligible. In the case of PC=50, the US gains 0.01 regardless of the tariff rate of 25% or 30%, while when China lowers the price to 45, the US gains 0.00 regardless of the tariff strategy.

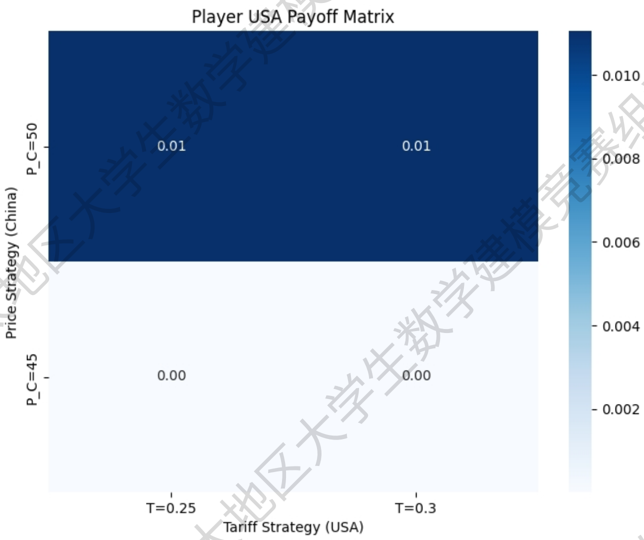

Figure 26 U.S. earnings matrix

As shown in Figure 27, the optimal strategy diagram of a dynamic game reveals the optimal strategy choice in a dynamic game. The red dots in the chart represent the optimal returns of China and the United States under the combination of PC=50 and T=0.25. This suggests that the optimal strategy for both sides is that China adopts a high price strategy while the United States maintains a low tariff level under the set parameters.

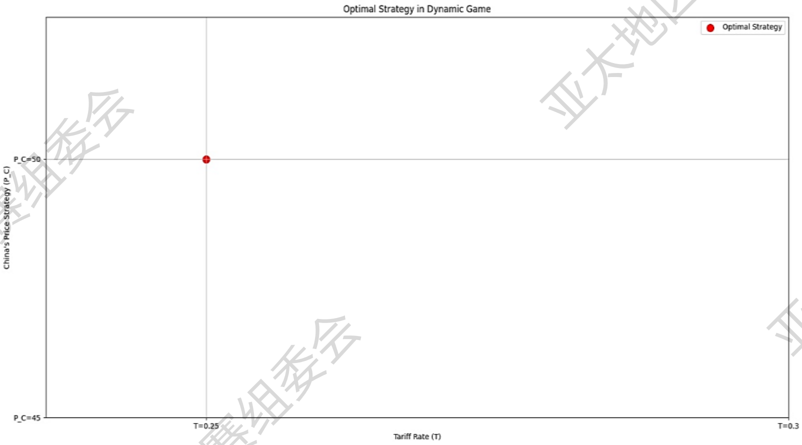

Fig.27 Dynamic game results

China's strategy in its export trade is to maintain high commodity pricing to ensure considerable gains even in the face of high tariff barriers. For the United States, the increase in tariffs is not effective in advancing its economic benefits, but may encourage China to stick to its high-price strategy, which in turn may weaken the competitiveness of American products in the market. According to the analysis results of the dynamic game, under certain strategic and parameter conditions, China has shown strong control in pricing, while the flexibility of the United States to adjust tariff policy has been limited.

IX、 Model evaluation

9. 1 Advantages of the model

(1) The Holt-Winters model effectively captures trends and seasonality in time series data, enhancing forecasting accuracy for datasets exhibiting these characteristics. Utilizing exponential smoothing methods, it autonomously adjusts weights to accommodate varying data patterns. This model is extensively applied across diverse domains, such as retail sales forecasting, economic analysis, and resource management.

(2) Static game theory models are relatively simple and easy to understand and analyze, especially when the game involves only a few participants and strategies. All participants make choices at the same time, and there is no precedence, which makes the model easier to handle in some cases. In a static game, the Nash equilibrium has good stability, and the players can unanimously predict the final form of the equilibrium.

(3) Dynamic game theory considers the sequence of actions and the gradual revelation of information, which is closer to the real-world decision-making process. Participants can adjust their strategies based on previous results and actions of other participants, which increases the flexibility and applicability of the model. For example, the perfect Nash equilibrium of sub-games provides a more refined concept of equilibrium to deal with credible threats and commitment problems in dynamic games.

9. 2 Disadvantages of the model

（1） The Holt-Winters method requires manual selection of smoothing coefficient and seasonal period, which may affect the accuracy of the prediction results. If the data contains irregular fluctuations, biased predictions can be generated.

（2） Static game theory does not take into account the time factor and the dynamic adjustment of strategies, which limits its application in real-world complex situations. Dynamic game-theoretic models are more complex and need to deal with multi-stage decision-making and information incompleteness, which increases the difficulty of analysis.

X、 References

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XI、 Appendix

Question one code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import Ridge, Lasso
from sklearn.preprocessing import StandardScaler
from scipy.stats import pearsonr
import matplotlib

matplotlib.rcParams['font.sans-serif'] = ['SimHei']
matplotlib.rcParams['axes.unicode_minus'] = False

file_path = 'D:/Desktop/Question 1 Data.xlsx'
df = pd.read_excel(file_path, sheet_name="China")

y_cats = df["Cat (10,000)"].values
y_dogs = df["Dog (10,000)"].values
features = [
    "Pet Market Size (US$ 100 million)", "Pet Food Expenditure (US$ 100 million)",
    "Veterinary Services Expenditure (US$ 100 million)", "Pet Household Penetration
Rate in China",
    "Total Fertility Rate (Female Fertility Per Capita Births)", "Urban Population",
    "Food Production Index (2014-2016 = 100)", "GDP per Capita (current US$)"
]
X = df[features]

X_cleaned = X.replace([np.inf, -np.inf], np.nan).dropna()
y_cats_cleaned = y_cats[:len(X_cleaned)]
y_dogs_cleaned = y_dogs[:len(X_cleaned)]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_cleaned)

def calculate_correlations(X, y, feature_names):
    correlations = {}
    for feature in feature_names:
        correlations[feature] = pearsonr(X[feature], y)[0]
```

```
    return correlations

correlations_cats = calculate_correlations(X_cleaned, y_cats_cleaned, features)
correlations_dogs = calculate_correlations(X_cleaned, y_dogs_cleaned, features)

def plot_correlation(correlations, title):
    correlations = pd.Series(correlations)
    plt.figure(figsize=(14, 8))
    correlations.plot(kind='bar', color='skyblue')
    plt.title(title, fontsize=18)
    plt.ylabel("Pearson Correlation", fontsize=14)
    plt.xlabel("Feature", fontsize=14)
    plt.xticks(rotation=90, fontsize=12)
    plt.yticks(fontsize=12)
    plt.grid(axis='y', linestyle='--', alpha=0.7)

plot_correlation(correlations_cats, "Pearson Correlation with Cat (10,000)")
plot_correlation(correlations_dogs, "Pearson Correlation with Dog (10,000)")

plt.show()_
```

Question two code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

uploaded_file_path = 'Question 2 data —V2.xlsx'
data = pd.ExcelFile(uploaded_file_path)

us_data = data.parse("US")
france_data = data.parse("France")
germany_data = data.parse("Germany")

pet_columns = ["cat", "dog"]
for country_data in [us_data, france_data, germany_data]:
    country_data[pet_columns] = country_data[pet_columns].apply(pd.to_numeric,
errors='coerce')

max_rows = min(len(us_data), len(france_data), len(germany_data))
us_data, france_data, germany_data = us_data.iloc[:max_rows],
```

```
france_data.iloc[:max_rows], germany_data.iloc[:max_rows]

    def calculate_growth_rate(data, columns):
        growth_rates = {column: ((data[column].iloc[1:] - data[column].iloc[:-1]) / data[column].iloc[:-1]) * 100 for column in columns}
        return growth_rates

    us_growth_rates = calculate_growth_rate(us_data, pet_columns)
    france_growth_rates = calculate_growth_rate(france_data, pet_columns)
    germany_growth_rates = calculate_growth_rate(germany_data, pet_columns)

    valid_years = us_data["pet/year"].iloc[1:].values
    us_growth_rates = {k: v[:len(valid_years)] for k, v in us_growth_rates.items()}
    france_growth_rates = {k: v[:len(valid_years)] for k, v in france_growth_rates.items()}
    germany_growth_rates = {k: v[:len(valid_years)] for k, v in germany_growth_rates.items()}

    def plot_growth_rates(growth_rates, country_name, pet_type):
        plt.figure(figsize=(10, 6))
        for pet in pet_type:
            plt.plot(valid_years, growth_rates[pet], marker='o', label=f'{pet.capitalize()} {country_name}')
        plt.title(f'{country_name} Pet Growth Rates')
        plt.xlabel('Year')
        plt.ylabel('Growth Rate (%)')
        plt.legend()
        plt.grid(True)
        plt.show()

    plot_growth_rates(us_growth_rates, 'US', pet_columns)
    plot_growth_rates(france_growth_rates, 'France', pet_columns)
    plot_growth_rates(germany_growth_rates, 'Germany', pet_columns)
```

Question three code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
from statsmodels.tsa.arima.model import ARIMA
```

```
import matplotlib

matplotlib.rcParams['font.sans-serif'] = ['SimHei']
matplotlib.rcParams['axes.unicode_minus'] = False

data_years = [2019, 2020, 2021, 2022, 2023]
food_production = [440.7, 727.3, 1554.0, 1508.0, 2793.0]
food_exports = [154.1, 9.8, 12.2, 24.7, 39.6]

plt.figure(figsize=(12, 8))

plt.subplot(2, 1, 1)
plt.plot(data_years, food_production, marker='o', label='Total value of pet food（RMB）
', color='blue')
plt.title('Total value of pet food production ', fontsize=16)
plt.ylabel('Total value of out-put（RMB，one hundred million）', fontsize=12)
plt.grid(True)
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(data_years, food_exports, marker='o', label=''Total export value of pet food
（Dollar）', color='green')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Total export value（Dollar，one hundred million）', fontsize=12)
plt.grid(True)
plt.legend()

plt.tight_layout()
plt.show()

import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.tsa.holtwinters import ExponentialSmoothing
import matplotlib.pyplot as plt
import seaborn as sns

plt.rcParams['font.sans-serif'] = ['SimHei']
```

```
plt.rcParams['axes.unicode_minus'] = False

data = pd.read_excel('D:/Desktop/Question 2 data —V2.xlsx')
```

Question four code

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Parameters
alpha = 1.5    # Price sensitivity
P_A = 55    # USA price
C_C, C_A = 30, 35    # Production costs for China and USA
T_values = [0.25, 0.30]    # Tariff rates (USA strategy)
P_C_values = [50, 45]    # China's price strategies

# Market share function
def market_share(P_C, P_A, alpha):
    S_C = 1 / (1 + np.exp(alpha * (P_C - P_A)))    # China's market share
    return S_C, 1 - S_C    # (China, USA market shares)

# Profit function
def profit(P_C, P_A, C_C, C_A, T, alpha):
    S_C, S_A = market_share(P_C, P_A, alpha)
    pi_C = (P_C - C_C - T * P_C) * S_C    # China's profit
    pi_A = (P_A - C_A) * S_A    # USA's profit
    return pi_C, pi_A

# Static game: calculate the payoff matrix
payoff_matrix = np.zeros((2, 2, 2))    # 2x2x2 matrix for China's and USA's strategies
for i, P_C in enumerate(P_C_values):
    for j, T in enumerate(T_values):
        pi_C, pi_A = profit(P_C, P_A, C_C, C_A, T, alpha)
        payoff_matrix[i, j, 0] = pi_C    # China's payoff
        payoff_matrix[i, j, 1] = pi_A    # USA's payoff

# Dynamic game: backward induction to find optimal strategy
def dynamic_game(P_C_values, T_values, P_A, C_C, C_A, alpha):
    best_payoff = -np.inf
    best_strategy = None
    for j, T in enumerate(T_values):
```

```
        for i, P_C in enumerate(P_C_values):
            pi_C, pi_A = profit(P_C, P_A, C_C, C_A, T, alpha)
            if pi_A > best_payoff:    # USA maximizes its own payoff
                best_payoff = pi_A
                best_strategy = (P_C, T)
    return best_strategy

# Compute optimal strategy
optimal_strategy = dynamic_game(P_C_values, T_values, P_A, C_C, C_A, alpha)

# Visualization functions
def plot_payoff_matrix(payoff_matrix, P_C_values, T_values, player):
    # Extract payoffs for the selected player
    payoffs = payoff_matrix[:, :, player]

    plt.figure(figsize=(8, 6))
    sns.heatmap(payoffs, annot=True, fmt=".2f", cmap="Blues",
                xticklabels=[f"T={t}" for t in T_values],
                yticklabels=[f"P_C={p}" for p in P_C_values])
    plt.title(f"Payoff Matrix for Player {player+1}")
    plt.xlabel("USA Strategies (Tariff Rates)")
    plt.ylabel("China Strategies (Price)")
    plt.show()

# Plot the payoff matrices for both players
plot_payoff_matrix(payoff_matrix, P_C_values, T_values, 0)    # China's payoff matrix
plot_payoff_matrix(payoff_matrix, P_C_values, T_values, 1)    # USA's payoff matrix

# Print the optimal strategy
print(f"Optimal Strategy: China's Price = {optimal_strategy[0]}, USA's Tariff =
{optimal_strategy[1]}")
```